MAH, Commercialization Frictions, and Original-Drug Innovation

Abstract

This note develops a theoretical mechanism model and mechanism-calibration blueprint for how China's Marketing Authorization Holder (MAH) system can affect original-drug innovation. The model keeps the institutional timing explicit. Firms choose original-drug R&D intensity before successful drug opportunities are realized. If a successful opportunity is realized, the firm chooses a commercialization route: internal production, entrusted production while retaining marketing authorization, non-retained transfer/sale/out-licensing, or abandonment. MAH enters the model in two linked ways. First, it makes the MAH-style entrusted-production route part of the relevant commercialization choice set. Second, conditional on this route being institutionally available, MAH lowers the policy and contractual friction of using a qualified entrusted manufacturer while retaining authorization. Demand-side conditions enter only through a single reduced-form market-return shifter, which is held fixed in the main MAH comparative statics. The reform does not directly raise scientific productivity, lower clinical-trial cost, increase demand, eliminate residual holder-side monitoring burdens, or make entrusted production universally better than internal production or transfer. Instead, it raises the private option value of a successful opportunity for firms for which the retained external route is relevant. This higher expected future commercialization value enters the Bellman equation and raises current R&D intensity in the model. The note does not claim full structural identification or a complete causal empirical design. Instead, it maps model objects to feasible moments, states required normalizations, and explains how approval-side realized-product data can discipline composite objects such as the entrusted-production wedge, the innovation-arrival scale, and, in future modules, the entry-cost distribution.

Keywords: MAH; original-drug innovation; commercialization frictions; entrusted production; route choice; firm entry.
JEL codes: D22; L65; O31; O38; I18.

Contents

1 Introduction

The purpose of this note is to discipline a theoretical mechanism and a mechanism-calibration blueprint rather than to estimate a full dynamic structural model of the Chinese pharmaceutical industry. The central question is whether MAH can raise original-drug innovation by improving downstream commercialization options. The mechanism is an expected-profit channel. A firm chooses R&D before knowing how many drug opportunities will be successful, but it anticipates the institutional regime under which a successful opportunity will later be produced, entrusted, transferred, or abandoned. The empirical role of the model is correspondingly modest: it organizes which data moments are close to the mechanism and which objects those moments can discipline.

This logic follows two lines of literature. First, pharmaceutical innovation responds to expected downstream rewards. Market-size and expected-return models show that firms direct innovation toward projects with higher expected commercial value (Schmookler, 1966; Acemoglu and Linn, 2004; Finkelstein, 2004; Blume-Kohout and Sood, 2013; Dubois et al., 2015; Budish et al., 2015; Dranove et al., 2014). Second, the value of innovation depends on complementary assets and commercialization strategy. An innovator without manufacturing or distribution capability may need to cooperate with, license to, or sell to firms that possess those assets (Teece, 1986; Arora et al., 2001; Gans and Stern, 2003). MAH fits this logic because it changes the commercialization environment by allowing the holder to retain authorization while using a qualified external manufacturer.

The model is deliberately narrow. It does not treat MAH as a demand shock. Demand enters only through a single reduced-form market-return shifter for the Chinese original-drug environment. The model does not index R&D effort, opportunity arrival, route choice, or retained stock by therapeutic area. It does not assume that MAH makes science easier. It does not assume that entrusted production dominates internal production. It also does not track every product in a firm's portfolio. Instead, it follows the product-line innovation tradition by using a stock of products or certificates as a low-dimensional state and an innovation-arrival function linking R&D effort to expected new drug opportunities (Klette and Kortum, 2004). The resulting framework is suitable for a setting where the currently feasible evidence is closer to approval-side realized products, holder-producer links, and route-use proxies than to latent idea arrivals, project-level R&D expenditure, or project-specific success probabilities. Throughout the paper, original-drug approvals and retained certificates are treated as data-side realizations or proxies of the model's upstream opportunity-arrival objects, not as literal observations of latent ideas.

The model should be read as a partial-equilibrium mechanism model rather than a full dynamic structural model of the pharmaceutical industry. The baseline takes the market-return shifter R_t, the contract-manufacturing service price p_m, and the distributions of incumbent and potential entrant characteristics as given. MAH affects the firm problem through route-set expansion and a reduction in the MAH-reducible policy and contractual friction τ^E of the retained external route, while residual holder-side monitoring and quality-control burdens remain. The main theoretical results are comparative statics for private opportunity value, R&D intensity, and expected original-drug counts. The calibration section maps feasible data moments to composite objects, but it does not claim to identify all primitives or to solve a full general-equilibrium allocation.

2 Related Literature

The first relevant literature studies how expected rewards shape pharmaceutical innovation. Acemoglu and Linn (2004) show that potential market size affects entry of new drugs, and later work studies how insurance design, reimbursement, and expected profitability shift drug-development incentives (Finkelstein, 2004; Blume-Kohout and Sood, 2013; Dubois et al., 2015; Budish et al., 2015; Dranove et al., 2014). Recent work on China studies how regulatory and demand-side changes affect drug development (Jia et al., 2023; Barwick et al., 2026). This paper differs by focusing on a commercialization-friction channel rather than on market size, approval delay, or reimbursement.

The second literature studies commercialization strategy and complementary assets. Teece (1986) emphasizes that innovators' returns depend on access to complementary assets. Arora et al. (2001) and Gans and Stern (2003) analyze markets for technology and the choice between commercialization and cooperation with established firms. The MAH reform is naturally interpreted through this lens: it improves one way for an innovator to access manufacturing capability while retaining the authorization asset.

The third literature provides the modeling style. Klette and Kortum (2004) use stochastic innovation arrivals and product-line growth to connect firm-level R&D to aggregate innovation. The model below uses a simpler version of that idea: R&D intensity determines the expected number of original-drug opportunities, and retained opportunities enter a drug-stock state. The model avoids the full state space of dynamic industry models such as Ericson and Pakes (1995) and Bajari et al. (2007).

3 Institutional Background and Economic Timing

The model distinguishes legal registration timing from economic decision timing. Under China's *Provisions for Drug Registration*, drug registration includes applications for clinical trials, marketing authorization, renewal, and supplementary applications. After obtaining the Drug Approval License, the applicant becomes the MAH (National Medical Products Administration, 2022a). Therefore, the model should not say that a firm must already hold the final approval or a commercial manufacturing certificate before it starts R&D. The relevant point is forward-looking: expected future commercialization feasibility affects the value of current R&D.

The institutional content of MAH is more specific than a generic reform dummy. It changes who may legally hold marketing authorization, how production may be organized after authorization is obtained, and how authorization and production may be assigned across firms under a regulated relationship. The key institutional break is the separation of the authorization-holding role from a strict requirement of in-house production.[1] This separation is especially relevant for research-originating firms that can generate valuable projects but do not operate full internal manufacturing capacity.

Under the *Drug Administration Law*, an MAH may manufacture drug products by itself or entrust a qualified drug manufacturer. If production is contracted out, the MAH and the contract manufacturer must sign entrustment and quality agreements, and the MAH remains responsible for the drug's safety, efficacy, and quality (National Medical Products Administration, 2019).[2] The manufacturing supervision rules require the MAH to assess the contracted firm's quality assurance and risk-management capabilities, sign quality and entrustment agreements, and supervise performance (National Medical Products Administration, 2022b). Thus MAH does not remove regulation. It creates a regulated holder-producer separation route.

This point matters for the economic mechanism. Entrusted production under MAH is not an anonymous spot purchase of factory services. It is a legally structured authorization-production assignment in which the innovating holder may retain authorization while production is performed by a qualified outside manufacturer under contractual, quality-management, and supervisory requirements.[3] The holder does not hand off the project and walk away; it keeps legally meaningful lifecycle and quality-system responsibilities. The reform therefore lowers the policy and contractual friction of using this particular external route, without eliminating the residual monitoring and quality-control burden borne by the holder.

The economic timing is as follows. In period t, a firm chooses original-drug R&D intensity. In period $t+1$, successful original-drug opportunities may arrive. Conditional on an opportunity, the firm chooses among internal production, entrusted production while retaining MAH, non-retained transfer/sale/out-licensing, or abandonment. This timing

[1] The 2016 pilot plan authorized selected provinces and municipalities to pilot the MAH system and allowed eligible applicants to apply to become holders under the pilot arrangement; see General Office of the State Council of the People's Republic of China (2016). The former CFDA follow-up notice operationalized pilot implementation and supervision; see China Food and Drug Administration (2016).

[2] For the post-law responsibility architecture, see the 2022 NMPA provisions on MAH quality and safety responsibility, released on December 29, 2022 and effective March 1, 2023 (National Medical Products Administration, 2022c).

[3] The 2023 NMPA announcement on MAH entrusted production further specifies licensing, quality-management, inspection, and supervision requirements for entrusted production; see National Medical Products Administration (2023).

captures the expectation mechanism: MAH facilitates expected future commercialization profits, and this expected future value motivates more current R&D. The reform is not modeled as a demand shock, a direct scientific-productivity shock, or a generic fixed-cost reduction for all firms. It is modeled as a route-specific change in the feasibility and friction of retaining authorization while using a qualified entrusted producer.

4 Model

The model combines four standard blocks rather than importing one complete canonical model. The demand block uses a CES differentiated-variety environment (Dixit and Stiglitz, 1977; Melitz, 2003). The market-return interpretation follows the pharmaceutical innovation literature in which expected market size, revenue, or profit shapes drug R&D incentives (Acemoglu and Linn, 2004; Dubois et al., 2015; Barwick et al., 2026). The innovation-stock and R&D-arrival structure follows product-line innovation models (Klette and Kortum, 2004). The commercialization-route choice follows the markets-for-technology literature, where innovators choose between internal commercialization, cooperation, licensing, transfer, or other external routes (Arora et al., 2001; Gans and Stern, 2003). The contribution here is to adapt these blocks to the MAH setting, where the key institutional change is a regulated holder-producer separation route.

4.1 Scope and partial-equilibrium closure

The baseline model is a partial-equilibrium dynamic route-choice model. It takes as given the market-return environment, the contract-manufacturing service price, the distribution of firm characteristics, the potential entrant pool, and the institutional regime. Firms choose original-drug R&D intensity before successful opportunities are realized. Conditional on a successful opportunity, they choose a commercialization route.

Formally, a baseline partial-equilibrium MAH environment consists of

$$\mathcal{E}_t = \Big(R_t,\ p_{m,t},\ H(a,k,S,\tau^T,\mu^E),\ G_N(a,k,S,\tau^T,\mu^E),\ M_t,\ \tau^E(M_t)\Big), \tag{1}$$

where R_t is the market-return shifter, $p_{m,t}$ is the direct price or payment for qualified contract manufacturing, H is the distribution of incumbent firm characteristics, G_N is the distribution of potential entrant characteristics, M_t is the institutional regime, and $\tau^E(M_t)$ is the MAH-reducible policy and contractual friction of the entrusted route.

The model does not solve for product-market clearing, input-market clearing, a contract-manufacturing market-clearing price, or an invariant industry distribution in the baseline. Its purpose is to characterize how MAH-induced route-set expansion and entrusted-route friction reduction affect the private opportunity value Γ_i, R&D intensity

x_i^*, and expected original-drug counts λ_i^*.

4.2 Demand side and single-market return

The model begins with a demand block that microfounds the single market-return object used in the subsequent commercialization-route problem. The baseline model keeps a single market-return shifter rather than introducing a therapeutic-area state space. This is a deliberate modeling boundary. The paper studies how MAH changes commercialization options conditional on the expected market return to a successful original-drug opportunity. It does not model different therapeutic markets as separate firm-level innovation problems.

Consider the Chinese original-drug market in period t. A representative payer or consumer allocates original-drug expenditure E_t across differentiated original-drug varieties $\omega \in \Omega_t$. Preferences over these varieties are CES, following the standard monopolistic-competition demand system in Dixit and Stiglitz (1977) and its modern differentiated-variety use in firm-level models such as Melitz (2003):

$$X_t = \left[\int_{\omega\in\Omega_t} (q_{\omega t}x_{\omega t})^{\frac{\sigma-1}{\sigma}}\,\mathrm{d}\omega\right]^{\frac{\sigma}{\sigma-1}}, \qquad \sigma > 1, \tag{2}$$

where $x_{\omega t}$ is quantity and $q_{\omega t}$ is a quality, novelty, or therapeutic-value shifter. The expenditure constraint is

$$\int_{\omega\in\Omega_t} p_{\omega t}x_{\omega t}\mathrm{d}\omega = E_t. \tag{3}$$

The associated quality-adjusted CES price index is

$$P_t = \left[\int_{\omega\in\Omega_t} q_{\omega t}^{\sigma-1}p_{\omega t}^{1-\sigma}\mathrm{d}\omega\right]^{\frac{1}{1-\sigma}}. \tag{4}$$

Demand for variety ω is therefore

$$x_{\omega t} = E_t P_t^{\sigma-1} q_{\omega t}^{\sigma-1} p_{\omega t}^{-\sigma}, \tag{5}$$

and revenue is

$$p_{\omega t}x_{\omega t} = E_t P_t^{\sigma-1} q_{\omega t}^{\sigma-1} p_{\omega t}^{1-\sigma}. \tag{6}$$

If a marketed original drug has marginal operating cost $c_{\omega t}$ and is priced with the CES markup

$$p_{\omega t} = \frac{\sigma}{\sigma-1}c_{\omega t}, \tag{7}$$

then per-period operating profit is

$$\varpi_{\omega t} = (p_{\omega t} - c_{\omega t})\,x_{\omega t} = \frac{1}{\sigma}p_{\omega t}x_{\omega t} = \frac{1}{\sigma}E_t P_t^{\sigma-1} q_{\omega t}^{\sigma-1} p_{\omega t}^{1-\sigma}. \tag{8}$$

Following the pharmaceutical innovation literature that links R&D incentives to expected market rewards (Acemoglu and Linn, 2004; Dubois et al., 2015; Barwick et al., 2026), the main model does not track product-specific demand or firm-product revenue. It compresses the expected operating-profit component of a successful original-drug opportunity into the single market-return shifter

$$R_t = \chi_t \mathbb{E}_\omega[\varpi_{\omega t}], \qquad \chi_t > 0, \tag{9}$$

The object R_t is not the full lifetime value of a retained drug. It is the route-period commercialization-event payoff associated with converting a successful original-drug opportunity into a marketed product. It excludes the continuation value of retaining the product in the innovating firm's future portfolio. That continuation value is represented separately by v_t.

The multiplier χ_t should therefore be interpreted as a reduced-form mapping from operating-profit conditions into the commercialization-event payoff, not as a full product-lifetime multiplier. Equivalently, R_t is increasing in aggregate demand-side conditions such as original-drug expenditure, patient population, disease burden, reimbursement environment, procurement exposure, and the quality or novelty composition of successful opportunities. Demand-side conditions discipline the background level of R_t, but MAH is not modeled as shifting R_t in the baseline comparative statics.

The demand block does not say that MAH changes demand. The CES block microfounds the market-return shifter R_t. It does not make the baseline model a product-market general-equilibrium model. In particular, the main MAH comparative statics hold R_t, E_t, and P_t fixed. MAH affects innovation through route-set expansion and entrusted-route friction reduction. Feedback from MAH-induced innovation into the price index, aggregate expenditure allocation, or product-market competition would require a separate equilibrium extension.

A more granular model could index opportunities by therapeutic area, but doing so would require therapeutic-area-specific R&D effort, opportunity arrival, retained-product stock, and route choice. The present paper does not take that step. Therapeutic-area information can still enter the empirical work as a control or composition adjustment for R_t.

4.3 Firm characteristics and state variables

Each firm i has fixed research and manufacturing abilities,

$$a_i > 0, \qquad k_i > 0, \tag{10}$$

where a_i is research ability and k_i is manufacturing ability. The joint distribution of firm characteristics is unrestricted. Research ability and manufacturing ability may be positively correlated, negatively correlated, or independent. This is important because the model should not impose that research-oriented firms always lack manufacturing capability.

The firm also has fixed commercialization characteristics (S_i, τ_i^T, μ_i^E). The object $S_i - \tau_i^T$ is the net non-retained transfer outside option defined below. The new object $\mu_i^E \geq 0$ is the residual holder-side burden of using entrusted production after MAH makes the retained external route feasible: quality-control work, technical transfer, producer supervision, and cross-firm coordination that remain with the holder. This burden is not lowered mechanically by MAH in the baseline. The full incumbent distribution is therefore $H(a, k, S, \tau^T, \mu^E)$.

The endogenous state is

$$n_{it} \geq 0, \tag{11}$$

where n_{it} is the stock of retained original drugs or original-drug certificates held by firm i at the beginning of period t. Existing retained drugs generate flow payoff

$$\pi n_{it}, \qquad \pi > 0. \tag{12}$$

The stock depreciates, expires, or becomes obsolete at rate $\delta \in (0, 1)$. This product-stock state is a low-dimensional substitute for a full multi-product portfolio, following product-line innovation models (Klette and Kortum, 2004).

4.4 R&D production function and uncertainty

The firm chooses original-drug R&D intensity

$$x_{it} \geq 0. \tag{13}$$

R&D intensity produces a random number of successful original-drug opportunities in the next period. Let $Y_{i,t+1}$ denote the number of such opportunities that reach the commercialization-route decision stage. The object $Y_{i,t+1}$ is not a raw scientific idea arrival at the laboratory stage. In approval-side data, approvals, retained certificates, or realized original-drug records are downstream proxies for this object. A richer pipeline model could introduce an earlier idea-arrival stage and a separate clinical or regulatory success probability, but the baseline mechanism model absorbs those probabilities into effective research ability a_i. The drug-count production function is

$$\lambda_{i,t+1}(x_{it}) \equiv \mathbb{E}_t[Y_{i,t+1}] = f(a_i, x_{it}) = a_i x_{it}. \tag{14}$$

Equation (14) is the model's production function for the number of new original-drug opportunities or certificates. It is not physical pill production. It says that research effort raises the expected number of successful opportunities, and higher research ability makes effort more productive. This is the simplest version of the stochastic innovation-arrival structure used in product-line innovation models (Klette and Kortum, 2004). A Poisson count interpretation is optional,

$$Y_{i,t+1} \mid x_{it}, a_i \sim \text{Poisson}(a_i x_{it}), \tag{15}$$

but the comparative statics below require only the conditional mean in equation (14). Any unobserved scientific success probability can be absorbed into the effective research ability a_i.

R&D effort has convex cost,

$$C(x_{it}) = \frac{\kappa}{2} x_{it}^2, \qquad \kappa > 0. \tag{16}$$

This reduced-form cost delivers an interior closed-form solution. It is not a claim that project-level R&D spending is observed. It is a tractability assumption used to make the expected-return channel transparent.

4.5 MAH indicator, route availability, and entrusted-production friction

Let $M_{t+1} \in \{0, 1\}$ denote the institutional regime relevant for commercialization in period $t + 1$. The variable $M_{t+1} = 0$ denotes the pre-MAH regime, and $M_{t+1} = 1$ denotes the MAH regime. The MAH-style entrusted-production route is available only when $M_{t+1} = 1$. The feasible route set is

$$\mathcal{R}(M_{t+1}) = \{I, T, A\} \cup \{E : M_{t+1} = 1\}, \tag{17}$$

where I is internal production, E is entrusted production while retaining MAH, T is a non-retained outside option such as transfer, sale, or out-licensing that removes the opportunity from the innovating firm's retained stock, and A is abandonment or delay. This discrete route set follows the commercialization-strategy logic in which innovators choose between internal development and external technology-market arrangements depending on the availability of a market for technology or ideas (Arora et al., 2001; Gans and Stern, 2003).

Conditional on the entrusted route being available, MAH also lowers the policy and contractual friction of using that route. A convenient way to write this is

$$\tau^E_{i,t+1}(M_{t+1}) = \bar{\tau}^E_i - \Delta\tau_i M_{t+1}, \qquad \Delta\tau_i \geq 0. \tag{18}$$

Equation (18) should be read together with the route set in equation (17). When $M_{t+1} = 0$, the MAH-style entrusted route is not feasible for the innovating holder. When $M_{t+1} = 1$, the route is feasible and its friction is lower. This formulation captures both ideas: MAH adds a regulated holder-producer separation option, and the strength of the option is governed by the friction τ^E.

The object τ^E is deliberately route-specific and MAH-reducible. It is not the market return R, the research productivity parameter a_i, the R&D cost parameter κ, the direct payment for entrusted manufacturing services p_m, or the residual holder-side burden μ^E_i. It captures institutional uncertainty, contractual responsibility allocation, and regulatory-compliance ambiguity associated with retaining authorization while using a qualified producer. A fall in τ^E means that the external route becomes more legally supported, contractually structured, and regulatorily intelligible. It does not mean that the holder has no responsibility or that physical manufacturing, monitoring, technical transfer, and quality control become free.

For derivative comparative statics, it is useful to study Γ as a function of the continuous friction τ^E conditional on E being feasible. The binary reform effect can then be decomposed into route-set expansion plus a fall in τ^E.

4.6 Commercialization route choice in period $t + 1$

For each successful original-drug opportunity, the firm chooses one route $r \in \mathcal{R}(M_{t+1})$. Let R_{t+1} be the current commercialization-event payoff from a commercialized original drug in period $t+1$. This is the single-market reduced-form return shifter from equation (9). It summarizes the non-stock payoff associated with converting a successful opportunity into a marketed product, including market size, therapeutic value, reimbursement environment, and product demand. It is included because pharmaceutical innovation responds to expected market rewards (Acemoglu and Linn, 2004; Dubois et al., 2015; Barwick et al., 2026). The model does not require firm-product revenue data.

Payoff accounting. There are three distinct payoff objects. First, R_t is the commercialization-event payoff from turning a successful opportunity into a marketed product. Second, existing retained products generate the flow payoff πn_{it}. Third, v_t is the marginal continuation value of adding one retained original-drug asset to the firm's stock. Retained routes receive v_t; non-retained routes do not. Thus $R_t \neq v_t$ and $R_t \neq \pi n_{it}$.

Assumption 1 (Payoff accounting and route ownership)**.** *R_{t+1} is the route-period commercialization payoff and excludes the continuation value of keeping the product in the innovating firm's retained stock. That continuation value is represented separately by $v_{t+1}\rho^r$. The route T is a non-retained outside option: after transfer, sale, or non-retained out-licensing, the innovating firm receives $S_i - \tau_i^T$ but does not add the product to its retained stock. A retained licensing arrangement would be a different route and is not modeled separately here.*

Non-retained transfer option. The route T is a non-retained outside option. It includes transfer, sale, or non-retained out-licensing arrangements under which the innovating firm receives a net payment but does not add the product to its retained stock. We write this payoff as

$$b_i^T = S_i - \tau_i^T. \tag{19}$$

The object S_i is the gross expected non-retained transfer value, and τ_i^T is the transaction or contracting friction associated with that outside option. In the baseline, $S_i - \tau_i^T$ is taken as exogenous. A useful special case is

$$S_i = \omega_i^T R_t, \qquad \omega_i^T \in [0, 1], \tag{20}$$

but the main comparative statics do not require imposing this functional form.

Current route payoffs in period $t + 1$ are

$$b_{i,t+1}^I = R_{t+1} - C^I(k_i), \tag{21}$$

$$b_{i,t+1}^E = R_{t+1} - p_{m,t+1} - \tau_{i,t+1}^E(M_{t+1}) - \mu_i^E, \tag{22}$$

$$b_{i,t+1}^T = S_i - \tau_i^T, \tag{23}$$

$$b_{i,t+1}^A = 0. \tag{24}$$

Here $C^I(k_i)$ is internal production cost, with

$$\frac{\partial C^I(k_i)}{\partial k_i} < 0, \tag{25}$$

so higher manufacturing ability lowers internal production cost. Here $p_{m,t+1}$ is the direct price or payment for qualified entrusted manufacturing services. The separate object $\tau_{i,t+1}^E$ captures the MAH-reducible policy and contractual friction of the retained external route. The residual burden μ_i^E captures holder-side monitoring, quality-control, technical-transfer, and coordination costs that remain even when the route is legally supported. In the baseline partial-equilibrium model p_m is taken as given. A later extension may determine p_m from a contract-manufacturing service market-clearing condition. $S_i - \tau_i^T$ is the net value of the non-retained outside option.

A convenient example is

$$C^I(k_i) = F_I + \frac{\bar{c}_I}{k_i}, \qquad k_i > 0, \tag{26}$$

where F_I is the fixed internal commercialization burden and $\bar{c}_I/k_i$ captures the idea that higher manufacturing capability lowers the cost of internal production. The main comparative statics require only $C^I_k(k_i) < 0$; the closed-form results do not depend on this specific functional form.

If the firm chooses internal production or entrusted production, it retains the drug in its own future stock. If it transfers, sells, uses a non-retained out-license, or abandons, the innovating firm does not retain it. Define

$$\rho^I = \rho^E = 1, \qquad \rho^T = \rho^A = 0. \tag{27}$$

Given the marginal value v_{t+1} of one retained original drug at the beginning of period $t+1$, the route-specific value of a successful opportunity is

$$G^r_{i,t+1} = b^r_{i,t+1} + v_{t+1}\rho^r. \tag{28}$$

The firm chooses

$$r^*_{i,t+1} \in \arg\max_{r\in\mathcal{R}(M_{t+1})} G^r_{i,t+1}. \tag{29}$$

Define the expected value of one successful original-drug opportunity as

$$\Gamma_{i,t+1}(M_{t+1}, \tau^E_{i,t+1}, \mu^E_i) = \max_{r\in\mathcal{R}(M_{t+1})} \left\{ b^r_{i,t+1} + v_{t+1}\rho^r \right\}. \tag{30}$$

When $M_{t+1} = 1$, this becomes

$$\begin{aligned} \Gamma^1_{i,t+1}(\tau^E_{i,t+1}, \mu^E_i) = \max\{&R_{t+1} - C^I(k_i) + v_{t+1}, \\ &R_{t+1} - p_{m,t+1} - \tau^E_{i,t+1} - \mu^E_i + v_{t+1}, \\ &S_i - \tau^T_i,\ 0\}. \end{aligned} \tag{31}$$

When $M_{t+1} = 0$, the MAH-style entrusted route is not available:

$$\begin{aligned} \Gamma^0_{i,t+1} = \max\{&R_{t+1} - C^I(k_i) + v_{t+1}, \\ &S_i - \tau^T_i,\ 0\}. \end{aligned} \tag{32}$$

Equations (31) and (32) show both parts of the mechanism. MAH adds the entrusted-production route to the commercialization choice set. Conditional on that route being available, a lower τ^E increases its value, but the residual burden μ^E_i prevents the route from becoming costless. The model does not imply that entrusted production is always chosen.

4.7 Recursive Bellman equation

The state transition is

$$n_{i,t+1} = (1-\delta)n_{it} + \rho^*_{i,t+1} Y_{i,t+1}, \tag{33}$$

where $\rho^*_{i,t+1} = \rho^{r^*_{i,t+1}}$. Equation (33) is written in expected-count form. If one wants to track multiple realized opportunities one by one, the last term can be written as a summation over realized opportunities. That notation is not needed for the main model.

The recursive problem is

$$\begin{aligned} V_t(n_{it}; a_i, k_i, S_i, \tau^T_i, \mu^E_i, M_t) = \max_{x_{it}\geq 0} \Bigg\{ & \pi n_{it} - \frac{\kappa}{2} x^2_{it} \\ & + \beta \mathbb{E}_t \Big[V_{t+1}\left(n_{i,t+1}; a_i, k_i, S_i, \tau^T_i, \mu^E_i, M_{t+1}\right) \\ & \qquad + Y_{i,t+1} b^{r^*_{i,t+1}}_{i,t+1} \Big] \Bigg\}. \end{aligned} \tag{34}$$

This equation shows the mechanism directly. Current R&D effort x_{it} raises the expected number of next-period successful opportunities. Those opportunities are valued through the next-period route decision, which is affected by MAH through both $\mathcal{R}(M_{t+1})$ and $\tau^E_{i,t+1}(M_{t+1})$, while the residual entrusted-route burden μ^E_i remains a firm-specific cost. Thus expected future commercialization profits enter the current R&D choice, as in product-line innovation models where current investment responds to the continuation value of future product opportunities (Klette and Kortum, 2004).

Because the value function is affine in the stock state, write

$$V_t(n_{it}; \cdot) = A_t(\cdot) + v_t n_{it}. \tag{35}$$

The marginal value of one retained original drug satisfies

$$v_t = \pi + \beta(1-\delta)v_{t+1}. \tag{36}$$

Using equations (14), (30), and (36), the Bellman reduces to

$$\begin{aligned} V_t(n_{it}; \cdot) = {} & \pi n_{it} + \beta A_{t+1} + \beta(1-\delta) v_{t+1} n_{it} \\ & + \max_{x_{it}\geq 0} \left\{ \beta a_i x_{it} \mathbb{E}_t \left[\Gamma_{i,t+1}(M_{t+1}, \tau^E_{i,t+1}, \mu^E_i) \right] - \frac{\kappa}{2} x^2_{it} \right\}. \end{aligned} \tag{37}$$

The reduced problem keeps the same economics as the full Bellman but makes the expected-profit channel transparent.

5 Closed-Form Solution

The R&D choice solves

$$\max_{x_{it}\geq 0}\left\{\beta a_i x_{it}\mathbb{E}_t[\Gamma_{i,t+1}]-\frac{\kappa}{2}x_{it}^2\right\}. \tag{38}$$

The first-order condition is

$$\beta a_i\mathbb{E}_t[\Gamma_{i,t+1}]-\kappa x_{it}=0. \tag{39}$$

Since abandonment gives value zero, $\Gamma_{i,t+1}\geq 0$. The optimal R&D intensity is

$$x_{it}^*=\frac{\beta a_i}{\kappa}\mathbb{E}_t[\Gamma_{i,t+1}]. \tag{40}$$

The expected number of next-period successful original-drug opportunities is

$$\lambda_{i,t+1}^*\equiv\mathbb{E}_t[Y_{i,t+1}^*]=a_i x_{it}^*=\frac{\beta a_i^2}{\kappa}\mathbb{E}_t[\Gamma_{i,t+1}]. \tag{41}$$

Equations (40) and (41) give the mechanism in closed form: expected future commercialization value raises current R&D and expected future original-drug counts.

For stationary comparative statics, hold the market-return background, the contract-manufacturing service price, and the residual entrusted-route burden fixed and set $R_t=R$, $p_{m,t}=p_m$, $M_t=M$, $\tau_{it}^E=\tau^E$, $v_t=v$, and $A_t=A$. From equation (36),

$$v=\frac{\pi}{1-\beta(1-\delta)}. \tag{42}$$

The post-MAH opportunity value as a function of the continuous entrusted-production friction is

$$\Gamma_i^1(k_i,S_i,\tau_i^T,\tau_i^E,\mu_i^E)=\max\left\{R-C^I(k_i)+v,R-p_m-\tau_i^E-\mu_i^E+v,S_i-\tau_i^T,0\right\}. \tag{43}$$

The pre-MAH opportunity value is

$$\Gamma_i^0(k_i,S_i,\tau_i^T)=\max\left\{R-C^I(k_i)+v,S_i-\tau_i^T,0\right\}. \tag{44}$$

The stationary solutions under regime M are

$$x_i^*(M)=\frac{\beta a_i}{\kappa}\Gamma_i^M, \tag{45}$$

$$\lambda_i^*(M)=\frac{\beta a_i^2}{\kappa}\Gamma_i^M, \tag{46}$$

$$V_i(n_i;M)=\frac{\beta^2 a_i^2}{2\kappa(1-\beta)}\left[\Gamma_i^M\right]^2+\frac{\pi}{1-\beta(1-\delta)}n_i. \tag{47}$$

where $\Gamma_i^M=\Gamma_i^0(k_i,S_i,\tau_i^T)$ if $M=0$ and $\Gamma_i^M=\Gamma_i^1(k_i,S_i,\tau_i^T,\tau_i^E,\mu_i^E)$ if $M=1$.

6 Comparative Statics

The comparative statics hold the reduced-form market-return shifter fixed. The main result is not that a lower generic cost raises value. The main result is a sorting statement: MAH creates a retained external commercialization route, and a firm responds strictly only when that route beats the firm's best pre-existing alternative.

6.1 Best pre-existing alternative and retained external route

Define the firm's best pre-existing non-entrusted alternative as

$$B_i(k_i, S_i, \tau_i^T) = \max\left\{R - C^I(k_i) + v,\ S_i - \tau_i^T,\ 0\right\}, \tag{48}$$

and define the retained external route value, conditional on MAH route availability, as

$$E_i(1) = R - p_m - \tau_i^E(1) - \mu_i^E + v. \tag{49}$$

Then

$$\Gamma_i^0 = B_i(k_i, S_i, \tau_i^T), \qquad \Gamma_i^1(k_i, S_i, \tau_i^T, \tau_i^E, \mu_i^E) = \max\left\{B_i(k_i, S_i, \tau_i^T), E_i(1)\right\}. \tag{50}$$

The entrusted route is relevant for firm i when

$$E_i(1) > B_i(k_i, S_i, \tau_i^T). \tag{51}$$

This condition gives the economic content of heterogeneity. A firm with weaker internal manufacturing capability has a higher $C^I(k_i)$ and therefore a weaker internal-production fallback. A firm with a stronger non-retained transfer or sale option has a higher $S_i - \tau_i^T$ and may not need the entrusted route. A firm with a high residual burden μ_i^E may find entrusted production unattractive even after MAH. Research ability a_i does not determine route relevance by itself, but once the route raises Γ_i, it scales the R&D and count responses through equations (45) and (46).

Proposition 1 (Strict MAH response and sorting)**.** *The total MAH effect on firm i's opportunity value is*

$$\Delta\Gamma_i^{MAH} = \Gamma_i^1(k_i, S_i, \tau_i^T, \tau_i^E, \mu_i^E) - \Gamma_i^0(k_i, S_i, \tau_i^T) = \left[E_i(1) - B_i(k_i, S_i, \tau_i^T)\right]_+. \tag{52}$$

Therefore

$$\Delta x_i^* = \frac{\beta a_i}{\kappa}\left[E_i(1) - B_i(k_i, S_i, \tau_i^T)\right]_+, \qquad \Delta\lambda_i^* = \frac{\beta a_i^2}{\kappa}\left[E_i(1) - B_i(k_i, S_i, \tau_i^T)\right]_+. \tag{53}$$

The response is strict if and only if $E_i(1) > B_i(k_i, S_i, \tau_i^T)$. Equivalently,

$$\left[E_i(1) - B_i(k_i, S_i, \tau_i^T)\right]_+ = [E_i(1) - B_i]_+.$$

Proof. Equation (50) gives $\Gamma_i^0 = B_i$ and $\Gamma_i^1 = \max\{B_i, E_i(1)\}$. Therefore $\Gamma_i^1 - \Gamma_i^0 = \max\{B_i, E_i(1)\} - B_i = [E_i(1) - B_i]_+$. Subtracting equations (45) and (46) across regimes gives equation (53). □

Lemma 1 (Route-set expansion)**.** *For every firm i and every finite τ_i^E and μ_i^E,*

$$\Gamma_i^1(k_i, S_i, \tau_i^T, \tau_i^E, \mu_i^E) \geq \Gamma_i^0(k_i, S_i, \tau_i^T). \tag{54}$$

Proof. The pre-MAH value in equation (44) maximizes over $\{I, T, A\}$. The post-MAH value in equation (43) maximizes over the larger set $\{I, E, T, A\}$. The maximum over a superset cannot be lower. □

6.2 Manufacturing-capability cutoff

The model also implies a manufacturing-capability cutoff. The retained external route beats internal production when

$$R - p_m - \tau_i^E(1) - \mu_i^E + v > R - C^I(k_i) + v, \tag{55}$$

or equivalently

$$C^I(k_i) > p_m + \tau_i^E(1) + \mu_i^E. \tag{56}$$

Under the illustrative cost function in equation (26), this condition becomes

$$F_I + \frac{\bar{c}_I}{k_i} > p_m + \tau_i^E(1) + \mu_i^E. \tag{57}$$

If $p_m + \tau_i^E(1) + \mu_i^E > F_I$, the external retained route beats internal production for firms with

$$k_i < \frac{\bar{c}_I}{p_m + \tau_i^E(1) + \mu_i^E - F_I}. \tag{58}$$

If $p_m + \tau_i^E(1) + \mu_i^E \leq F_I$, the retained external route is more attractive than internal production for every finite positive k_i under this example. This manufacturing-capability cutoff is only an internal-production comparison. In either case, the route may still fail to be chosen if the non-retained transfer option $S_i - \tau_i^T$ is strong.

Lemma 2 (Derivative of opportunity value with respect to entrusted-production friction)**.** *Conditional on the entrusted route being feasible, the opportunity value is weakly* *decreasing in τ_i^E:*

$$\frac{\partial \Gamma_i^1(k_i, S_i, \tau_i^T, \tau_i^E, \mu_i^E)}{\partial \tau_i^E} \leq 0. \tag{59}$$

If entrusted production is the unique value-maximizing route, then

$$\frac{\partial \Gamma_i^1(k_i, S_i, \tau_i^T, \tau_i^E, \mu_i^E)}{\partial \tau_i^E} = -1. \tag{60}$$

If another route is strictly value-maximizing, then

$$\frac{\partial \Gamma_i^1(k_i, S_i, \tau_i^T, \tau_i^E, \mu_i^E)}{\partial \tau_i^E} = 0. \tag{61}$$

At ties, the subgradient lies in $[-1, 0]$.

Proof. From equation (43), only the entrusted-production term

$$R - p_m - \tau_i^E - \mu_i^E + v \tag{62}$$

depends on τ_i^E. The envelope theorem for the maximum operator gives equations (60) and (61). The subgradient statement follows from convex analysis of a pointwise maximum. □

Proposition 2 (Firm-level R&D response)**.** *Conditional on MAH route availability, optimal R&D intensity is weakly decreasing in the entrusted-production friction:*

$$\frac{\partial x_i^*}{\partial \tau_i^E} = \frac{\beta a_i}{\kappa} \frac{\partial \Gamma_i^1(k_i, S_i, \tau_i^T, \tau_i^E, \mu_i^E)}{\partial \tau_i^E} \leq 0. \tag{63}$$

Therefore a reduction in τ_i^E weakly raises x_i^*.

Proof. Differentiate equation (45) with respect to τ_i^E and apply Lemma 2. If entrusted production is uniquely optimal, $\partial x_i^* / \partial \tau_i^E = -\beta a_i / \kappa < 0$. □

Proposition 3 (Original-drug count response)**.** *The expected number of successful original-drug opportunities is weakly decreasing in the entrusted-production friction:*

$$\frac{\partial \lambda_i^*}{\partial \tau_i^E} = \frac{\beta a_i^2}{\kappa} \frac{\partial \Gamma_i^1(k_i, S_i, \tau_i^T, \tau_i^E, \mu_i^E)}{\partial \tau_i^E} \leq 0. \tag{64}$$

Therefore a reduction in τ_i^E weakly raises expected original-drug opportunities or certificates.

Proof. Differentiate equation (46) with respect to τ_i^E and apply Lemma 2. □

Proposition 4 (Binary MAH effect with route-set expansion and friction reduction)**.** *Let the post-MAH entrusted-production friction be*

$$\tau_i^E(M) = \bar{\tau}_i^E - \Delta\tau_i M, \qquad \Delta\tau_i \geq 0. \tag{65}$$

The total MAH effect on firm i's opportunity value can be written as

$$\begin{aligned} \Delta\Gamma_i^{MAH} = & \underbrace{\Gamma_i^1(k_i, S_i, \tau_i^T, \bar{\tau}_i^E, \mu_i^E) - \Gamma_i^0(k_i, S_i, \tau_i^T)}_{\text{route-set expansion}} \\ & + \underbrace{\left[\Gamma_i^1(k_i, S_i, \tau_i^T, \bar{\tau}_i^E - \Delta\tau_i, \mu_i^E) - \Gamma_i^1(k_i, S_i, \tau_i^T, \bar{\tau}_i^E, \mu_i^E)\right]}_{\text{friction reduction}} \geq 0. \end{aligned} \tag{66}$$

The first term is weakly nonnegative by Lemma 1. The second term is weakly nonnegative by Lemma 2.

Proof. Add and subtract $\Gamma_i^1(k_i, S_i, \tau_i^T, \bar{\tau}_i^E, \mu_i^E)$ from $\Gamma_i^1(k_i, S_i, \tau_i^T, \bar{\tau}_i^E - \Delta\tau_i, \mu_i^E) - \Gamma_i^0(k_i, S_i, \tau_i^T)$. Lemma 1 gives nonnegativity of the route-set term. Since $\bar{\tau}_i^E - \Delta\tau_i \leq \bar{\tau}_i^E$, Lemma 2 gives nonnegativity of the friction-reduction term. □

MAH does not mechanically raise innovation by lowering a generic friction. It creates a retained external commercialization route: an innovator can keep the authorization asset while using a qualified external producer. In the model, the innovation response is strict only when this retained external route exceeds the firm's best pre-existing alternative, namely internal production, non-retained transfer, or abandonment. Therefore, the strongest response should arise among high-research-capability firms with weak internal manufacturing fallback, weak non-retained transfer outside options, and manageable residual monitoring burdens.

Table 1: Model-Implied Differences Across Mechanisms

Mechanism	Model object	Route-use prediction	Innovation prediction
Demand expansion	$R\uparrow$	Does not specifically predict holder–producer split	Multiple commercialization routes become more attractive
Scientific productivity improvement	$a_i\uparrow$	Does not specifically predict entrusted-route use	Research-capable firms expand opportunities broadly
Review-speed or approval-probability reform	Lower delay or higher success probability	Does not necessarily change the retained external route	Applications or approvals can rise broadly
MAH route-friction channel	E route added, $\tau_i^E\downarrow$	Holder–producer split rises among relevant retained products	Strongest for high-a_i, low-k_i, weak transfer option, and low μ_i^E firms

These results are private option-value results. They do not imply that social welfare necessarily rises, that entrusted production is always chosen, or that every firm responds. The aggregate implications below are fixed-distribution sums of the firm-level threshold responses, not general-equilibrium innovation or welfare statements.

7 Aggregate Response Modules: Incumbents, Potential Entrants, and Expected Drug Counts

Let $H_I(a, k, S, \tau^T, \mu^E, n)$ be the distribution of incumbents and let $G_N(a, k, S, \tau^T, \mu^E)$ be the distribution of potential entrants. Let f^e be an entry cost with conditional cumulative distribution $F_e(\cdot \mid a, k, S, \tau^T, \mu^E)$ and density $f_e(\cdot \mid a, k, S, \tau^T, \mu^E)$.

The entry block is not a full free-entry equilibrium condition in the baseline model. It is a response module that maps the post-entry value $A(a, k, S, \tau^T, \mu^E, M)$ into potential research-oriented entry through an entry-cost distribution F_e. In the current approval-side data environment, this module should be interpreted as a future calibration or sensitivity exercise unless a credible firm-year entry panel is available.

The stationary entrant value, excluding old stock, is

$$A(a, k, S, \tau^T, \mu^E, M) = \frac{\beta^2 a^2}{2\kappa(1-\beta)} \left[\Gamma^M(a, k, S, \tau^T, \mu^E)\right]^2. \tag{67}$$

A potential entrant enters if $A(a, k, S, \tau^T, \mu^E, M) \geq f^e$. Therefore the mass of entrants is

$$N_E(M) = \int F_e\Big(A(a, k, S, \tau^T, \mu^E, M) \mid a, k, S, \tau^T, \mu^E\Big)\, \mathrm{d}G_N(a, k, S, \tau^T, \mu^E). \tag{68}$$

For a potential entrant with fixed $(a_i, k_i, S_i, \tau_i^T, \mu_i^E)$ and entry cost f_i^e, MAH strictly changes the entry decision only when

$$\begin{aligned} \Delta Entry_i > 0 \quad \Longleftrightarrow \quad & f_i^e \in [A(a_i, k_i, S_i, \tau_i^T, \mu_i^E, 0), \\ & A(a_i, k_i, S_i, \tau_i^T, \mu_i^E, 1)). \end{aligned} \tag{69}$$

Thus MAH does not make all potential entrants enter. It pushes across the entry threshold only those potential entrants whose entry costs lie between the pre-MAH and post-MAH operation values.

Conditional on MAH route availability, differentiating with respect to τ^E gives

$$\frac{\partial A(a, k, S, \tau^T, \mu^E, 1)}{\partial \tau^E} = \frac{\beta^2 a^2}{\kappa(1-\beta)} \Gamma^1(a, k, S, \tau^T, \mu^E, \tau^E) \frac{\partial \Gamma^1(a, k, S, \tau^T, \mu^E, \tau^E)}{\partial \tau^E} \leq 0. \tag{70}$$

Thus

$$\frac{\partial N_E(1)}{\partial \tau^E} = \int f_e\Big(A(a, k, S, \tau^T, \mu^E, 1) \mid a, k, S, \tau^T, \mu^E\Big) \frac{\partial A(a, k, S, \tau^T, \mu^E, 1)}{\partial \tau^E}\, \mathrm{d}G_N(a, k, S, \tau^T, \mu^E) \leq 0. \tag{71}$$

This is the standard entry logic that higher post-entry value attracts additional entrants (Bresnahan and Reiss, 1991; Berry, 1992), here applied to research-oriented entry.

Holding the distribution of incumbent characteristics fixed, the model-implied expected original-drug count from incumbents is

$$\Lambda_I(M) = \int \frac{\beta a^2}{\kappa} \Gamma^M(a, k, S, \tau^T, \mu^E) \,\mathrm{d}H_I(a, k, S, \tau^T, \mu^E, n). \tag{72}$$

Holding the potential entrant pool fixed, the model-implied expected original-drug count from entrants is

$$\Lambda_N(M) = \int F_e\Big(A(a, k, S, \tau^T, \mu^E, M) \mid a, k, S, \tau^T, \mu^E\Big) \frac{\beta a^2}{\kappa} \Gamma^M(a, k, S, \tau^T, \mu^E) \,\mathrm{d}G_N(a, k, S, \tau^T, \mu^E). \tag{73}$$

Holding the distribution of incumbent characteristics and the potential entrant pool fixed, the model-implied aggregate expected original-drug count is

$$\Lambda^{agg}(M) = \Lambda_I(M) + \Lambda_N(M). \tag{74}$$

This is an aggregate response under fixed distributions, not a general-equilibrium aggregate obtained from an invariant industry distribution. Conditional on MAH route availability, the incumbent-margin derivative is

$$\frac{\partial \Lambda_I(1)}{\partial \tau^E} = \int \frac{\beta a^2}{\kappa} \frac{\partial \Gamma^1(a, k, S, \tau^T, \mu^E, \tau^E)}{\partial \tau^E} \,\mathrm{d}H_I(a, k, S, \tau^T, \mu^E, n) \leq 0. \tag{75}$$

The entrant-margin derivative is

$$\begin{aligned} \frac{\partial \Lambda_N(1)}{\partial \tau^E} &= \int F_e\Big(A(a, k, S, \tau^T, \mu^E, 1) \mid a, k, S, \tau^T, \mu^E\Big) \frac{\beta a^2}{\kappa} \frac{\partial \Gamma^1(a, k, S, \tau^T, \mu^E, \tau^E)}{\partial \tau^E} \,\mathrm{d}G_N(a, k, S, \tau^T, \mu^E) \\ &\quad + \int \frac{\beta a^2}{\kappa} \Gamma^1(a, k, S, \tau^T, \mu^E, \tau^E) f_e\Big(A(a, k, S, \tau^T, \mu^E, 1) \mid a, k, S, \tau^T, \mu^E\Big) \frac{\partial A(a, k, S, \tau^T, \mu^E, 1)}{\partial \tau^E} \,\mathrm{d} \end{aligned} \tag{76}$$

Both terms are weakly negative by Lemma 2 and equation (70). Therefore,

$$\frac{\partial \Lambda^{agg}(1)}{\partial \tau^E} \leq 0. \tag{77}$$

A reduction in entrusted-production friction weakly increases the fixed-distribution aggregate response through both incumbent intensity and potential entrant response.

Corollary 1 (Aggregate fixed-distribution implication)**.** *For any fixed joint distribution of firm characteristics, and for any fixed conditional entry-cost distribution, MAH weakly raises model-implied aggregate expected original-drug opportunities through route-set expansion and friction reduction. Conditional on the entrusted route being feasible,*

$$\frac{\partial \Lambda^{agg}(1)}{\partial \tau^E} \leq 0, \tag{78}$$

so $\mathrm{d}\tau^E < 0$ *implies* $\mathrm{d}\Lambda^{agg} \geq 0$. *The increase is strict if a positive-measure set of incumbents or potential entrants has* $a > 0$, *positive operation or entry value, and the retained external route is relevant in the route-choice maximum.*

Proof. Equations (75) and (76) imply equation (77). Route-set expansion is weakly positive by Lemma 1. Strictness follows when $\partial\Gamma^1/\partial\tau^E < 0$ on a positive-measure set or when adding the entrusted route strictly raises the maximum for a positive-measure set. □

8 Mechanism Calibration Blueprint

The model is not intended to identify every primitive. The calibration exercise is a mechanism calibration: it asks whether data moments close to the route-choice channel can discipline a small number of composite objects. It is not a full structural estimation exercise and not a stand-alone causal identification design.

8.1 Model-to-data mapping

The key requirement is to keep model objects, empirical moments, and claim strength separate. Table 2 summarizes the mapping.

8.2 Executable calibration steps

A feasible blueprint has five steps.

Step 1: Define the sample and moments. Construct a comparable product-level sample of original-drug records. If available, also construct demand-side controls d_t^{data} for the single-market return shifter R_t, such as disease burden, patient population, reimbursement status, procurement exposure, or therapeutic-area composition. These controls are used to discipline or condition the market-return background, not to define MAH as a demand shock. The baseline approval-side moments are

$$m^{data} = \left(s_{post}^{E,data}, \bar{\lambda}_0^{O,data}, N_E^{data}\right),$$

where $s_{post}^{E,data}$ is the post-MAH holder–producer split share among comparable original-drug records, $\bar{\lambda}_0^{O,data}$ is a baseline realized original-drug count, and N_E^{data} is included only when a credible research-oriented entry panel is available. The demand-side control vector d_t^{data} can be reported alongside these moments or used in sensitivity checks for R_t.

Table 2: Model Objects, Data Moments, and Claim Boundaries

Model object	Economic meaning	Feasible data moment	Claim boundary
R_t	Single-market reduced-form commercialization return	Demand-side controls: disease burden, patient population, reimbursement, procurement exposure, aggregate sales or therapeutic-composition controls	Background/control for opportunity value, not an MAH treatment channel or a product-level demand estimate
p_m	Direct price or payment for qualified contract-manufacturing services	Contract-manufacturing prices, CMO fee schedules, cost reports, or sensitivity values if unavailable	Exogenous service price in the baseline; not part of the entrusted-route governance wedge
τ^E	MAH-reducible policy, contractual, and regulatory friction of the retained external route	Holder–producer split among comparable original-drug records; post-MAH exposure-gradient in split shares	Disciplines a composite route-use wedge, not legal cost, search cost, contract price, residual monitoring burden, or quality-management cost separately
μ_i^E	Residual holder-side monitoring, quality-control, technical-transfer, and coordination burden	Not separately identified in approval-side data; handled by heterogeneity, normalization, or sensitivity values	Prevents entrusted production from being treated as costless after MAH
Γ_i^M	Private expected value of a successful original-drug opportunity under regime M	Not directly observed; inferred through route-use and count moments after normalizations	Private option value, not social welfare
x_i^*	Original-drug R&D intensity	Usually unobserved in approval-side data; may be proxied by R&D spending, patents, or clinical-trial sponsorship in future modules	Not identified from approval records alone
λ_i^* and Y_{it}	Arrival rate and realized count of successful original-drug opportunities	Original-drug approvals, realized products, applications, or IND/CTA records, depending on data layer	Approval counts are downstream realizations, not latent idea arrivals
n_{it}	Stock of retained original drugs or certificates	Holder-level stock of retained original-drug approvals or certificates	Captures retained assets; transferred products require separate interpretation
N_E and F_e	Research-oriented entry and entry-cost distribution	First appearance in original-drug applications, clinical-trial sponsorship, or holder records	Future or sensitivity module unless firm-year entry data are merged

Step 2: Choose composite parameters. The calibration should target composite objects rather than primitive costs:

$$\theta = (\omega_E, \alpha_O, \mu_e)\,,$$

where ω_E denotes the composite entrusted-route wedge $(\tau^E + \mu^E)/\sigma_r$ relative to the route-choice scale, α_O denotes the composite innovation-arrival scale $\beta a^2/\kappa$, and μ_e indexes a parsimonious entry-cost distribution. In the baseline approval-side exercise, μ_e can be omitted or treated as a sensitivity parameter.

Step 3: State normalizations. Because approval-side data do not separately observe a_i, κ, Γ_i, project success probabilities, and route-choice scale, the calibration must normalize at least one object. Examples include $\kappa = 1$, a benchmark $\Gamma^0(a, k, S, \tau^T, \mu^E) = 1$, or a chosen route-choice scale σ_r. The calibrated parameters should be interpreted relative to these normalizations.

Step 4: Match moments. Let $m^{model}(\theta)$ denote the moments generated by the model under the chosen normalizations. The calibration can minimize

$$Q(\theta) = \left[m^{data} - m^{model}(\theta)\right]' W \left[m^{data} - m^{model}(\theta)\right], \tag{79}$$

where W is a transparent weighting matrix. With few moments, W can be diagonal and sensitivity to alternative weights should be reported rather than hidden.

Step 5: Report sensitivity and claim boundaries. The baseline report should show how the calibrated wedge and innovation-arrival scale change under alternative original-drug definitions, sample restrictions, holder-producer name cleaning rules, and route-use proxies. The exercise can support statements about consistency with the MAH commercialization-friction mechanism. It cannot separately estimate legal compliance costs, search costs, quality-agreement costs, contracting costs, contract-manufacturing prices, or firm-specific project success probabilities.

8.3 Route-use moments and logit smoothing

Route-share smoothing for calibration. The baseline theory uses deterministic route choice:

$$\Gamma_i^M = \max_{r \in \mathcal{R}(M)} G_i^r. \tag{80}$$

For route-share calibration only, introduce an idiosyncratic route-choice shock

$$\widetilde{G}_i^r = G_i^r + \varepsilon_i^r, \tag{81}$$

where ε_i^r is Type-I extreme value with scale parameter σ_r. This smoothing device is not needed for the comparative statics. It is used only to map observed route-use shares into an effective entrusted-route wedge.

The route-use block is a calibration device for the retained-product margin, not a complete route-choice estimation. To isolate the logic, compare internal production and entrusted production among successful opportunities that remain with the innovating holder. In that conditional comparison,

$$G_t^E - G_t^I = C^I(k) - p_{m,t} - \tau_t^E - \mu^E. \tag{82}$$

Conditional on comparing retained internal production and retained entrusted production,

$$P_i(r = E \mid r \in \{I, E\}) = \frac{\exp(G_i^E/\sigma_r)}{\exp(G_i^E/\sigma_r) + \exp(G_i^I/\sigma_r)}. \tag{83}$$

Therefore,

$$\log\left(\frac{s_t^E}{1 - s_t^E}\right) = \frac{C^I(k) - p_{m,t} - \tau_t^E - \mu^E}{\sigma_r}, \tag{84}$$

where s_t^E is a holder–producer split share among comparable retained original-drug records and σ_r is a route-choice scale. Without separate evidence on holder-side monitoring and quality-control costs, route-use moments discipline the composite effective wedge $(\tau_t^E + \mu^E)/\sigma_r$ relative to internal production costs and $p_{m,t}$.

The baseline should not mechanically use a pre-post log-odds difference when the pre-MAH entrusted route is structurally unavailable. In that case $s_{pre}^E = 0$ by institutional design, so $\log(s_{pre}^E/(1 - s_{pre}^E))$ is not a comparable route-use moment. The baseline interpretation should treat the reform as route-set expansion plus a conditional friction decline. If the data permit, post-MAH exposure variation provides the cleaner calibration margin:

$$\log\left(\frac{s_{\ell,post}^E}{1 - s_{\ell,post}^E}\right) = \eta_0 + \eta_1 Exposure_\ell + u_\ell, \tag{85}$$

where ℓ indexes an exposure cell, such as a region, dosage-form segment, firm type, or other implementation-intensity cell. The coefficient η_1 disciplines relative differences in the effective entrusted-route wedge across exposure cells. A pseudo-count for a zero pre-MAH split share can be reported only as a sensitivity device, not as the baseline economic interpretation.

8.4 Count and entry blocks

Baseline original-drug counts discipline the composite R&D arrival scale. From equation (46),

$$\bar{\lambda}_0^O = \mathbb{E}\left[\frac{\beta a^2}{\kappa}\Gamma^0(a, k, S, \tau^T, \mu^E)\right]. \tag{86}$$

If the data are approval-side realized products, this moment should be described as a realized-product count rather than a direct observation of latent idea arrival.

Entrant counts discipline the entry-cost distribution only when the data contain a credible firm-year measure of research-oriented entry. From equation (68),

$$N_E(M) = \int F_e\Big(A(a, k, S, \tau^T, \mu^E, M) \mid a, k, S, \tau^T, \mu^E\Big)\; \mathrm{d}G_N(a, k, S, \tau^T, \mu^E). \tag{87}$$

With only approval-side records, the entry block should be kept as a future module or sensitivity exercise.

8.5 Data requirements

The minimum product-level data are drug registration and approval records with product name, approval date, registration category, original-drug classification, holder, producer or manufacturer, dosage form, and indication or therapeutic class. These records are needed to construct original-drug counts and holder-producer split. Relevant official sources include NMPA and CDE registration and approval data, with careful cleaning of holder and producer names.

Demand-side data are useful as controls for the reduced-form market-return shifter R_t. They may include disease burden, patient population, reimbursement status, NRDL status, procurement exposure, aggregate therapeutic-area sales, prescription volume, or therapeutic-area composition. These variables help keep the MAH commercialization-friction channel distinct from changes in market size or payment environment. They should not be interpreted as evidence that MAH itself raises demand.

The minimum firm-level data are legal-entity identifiers, founding date, location, ownership, business scope, observed first drug project, and proxies for research and manufacturing ability. Research ability a_i can be proxied by pre-reform original-drug applications, patents, clinical trial sponsorship, or R&D expenditure where available. Manufacturing ability k_i can be proxied by Drug Manufacturing Certificates, GMP or manufacturing-site information, dosage-form scope, production license records, or manufacturing assets. Firm registries such as the National Enterprise Credit Information Publicity System and commercial databases such as Tianyancha or Qichacha can help standardize firm identities.

The route-use outcome should identify whether the MAH or holder differs from the actual producer. The main proxy is

$$Split_p = \mathbf{1}\{Holder_p \neq Producer_p\}. \tag{88}$$

This proxy is not the policy treatment. It is an observed route-use proxy for realized products, and it should be validated against name standardization, group affiliation, and product-level regulatory records where possible. Data on quality agreements, technology-transfer requirements, audits, or holder-manufacturer coordination would be useful for separating τ^E from μ^E, but the baseline approval-side data should not be interpreted as doing that separation.

Useful additional data include CDE clinical trial or IND/CTA records, CNIPA patent data, centralized procurement data, hospital procurement or sales databases where accessible, and regional policy exposure measures. These data help separate the MAH commercialization channel from broader market-size or demand-side changes, but they are not required for the theoretical mechanism itself.

8.6 Preferred eight-moment extension

The baseline mechanism calibration uses three moment families: route use, original-drug counts, and research-oriented entry. A richer implementation can expand these into eight empirical moments:

1. Original-drug holder–producer split share:

$$m_1 = \Pr(Split_p = 1 \mid O_p = 1).$$

   This disciplines the composite entrusted-route wedge relative to the route-choice scale.

2. Realized original-drug count:

$$m_2 = \mathbb{E}[Y_{it}^O].$$

   This disciplines the composite innovation-arrival scale.

3. Original-drug share among all realized products:

$$m_3 = \frac{Y_{it}^O}{Y_{it}^{all}}.$$

   This disciplines the original-drug orientation of realized innovation.

4. External-route use among research-oriented or B-like firms:

$$m_4 = \Pr(Split_p = 1 \mid O_p = 1,\ B\text{-like firm}).$$

This tests whether the mechanism is strongest for firms most likely to need external commercialization.

5. Application-to-approval conversion for original drugs:

$$m_5 = \frac{\#\ \text{Approved original drugs}}{\#\ \text{Original-drug applications}}.$$

This moment should be used only after application and status data are available.

6. Research-oriented entry:

$$m_6 = \#\{\text{first-time original-drug applicants or sponsors}\}.$$

This disciplines the entry-cost distribution only when a credible firm-year entry panel is available.

7. Heterogeneity by pre-reform manufacturing capability:

$$m_7 = \Delta Y^O_{LowManu} - \Delta Y^O_{HighManu}$$

or

$$m_7 = \Delta Split^O_{LowManu} - \Delta Split^O_{HighManu}.$$

This checks whether the response is stronger among firms with weaker internal manufacturing capacity.

8. Producer-side entrusted-manufacturing response:

$$m_8 = \#\{\text{qualified entrusted producers}\}$$

or a related measure of producer-side participation. This is needed for the optional contract-manufacturing market extension.

These moments are not all available in the current approval-side data. The current baseline should use feasible approval-side moments and reserve conversion, entry, and producer-side moments for future merged data modules.

9 Conclusion

The model gives a simple mechanism and a disciplined calibration blueprint. MAH adds the regulated route through which an innovating holder can retain marketing authorization while entrusting production to a qualified manufacturer. Conditional on the route being available and holding the reduced-form market-return background fixed, a lower MAH-reducible route friction raises the private route-option value Γ only for firms for which the retained external route beats the best pre-existing alternative. Because expected next-period route value enters the Bellman, current R&D intensity rises for those firms. Since the expected number of new original-drug opportunities is produced by $\lambda = ax$, higher R&D intensity raises expected original-drug counts. The calibration blueprint maps this mechanism to approval-side realized-product moments while preserving the boundary between latent opportunities, retained certificates, route-use proxies, demand-side controls, and realized approvals. The paper does not require project-level R&D investment, observed project-specific success probabilities, a complete causal design, product-level demand estimation, or a full equilibrium model of contract manufacturing.

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10 Appendix

A Institutional Details and Mechanism Clarification

This appendix expands the institutional discussion behind the route-specific friction in the model. The objective is not to claim that MAH simply removes regulation. The point is more precise: MAH creates a legally recognized route under which marketing authorization can remain with the holder while production is performed by a qualified outside manufacturer, and it pairs that route with explicit holder-side responsibility, contractual requirements, quality-management obligations, and supervisory expectations.

A.1 Institutional evolution from pilot to legal consolidation

The institutional evolution matters because MAH is not a loose industry slogan. It is a progressively formalized legal arrangement. The reform path first weakened the tight practical bundling of research, authorization, and in-house production by allowing authorization-holding and production to be separated under regulated conditions (General Office of the State Council of the People's Republic of China, 2016; China Food and Drug Administration, 2016). The later legal architecture then made the holder the entity associated with the drug registration certificate and located lifecycle responsibility on that holder (National Medical Products Administration, 2019, 2022c).

The key legal consolidation is that the holder may either manufacture by itself or entrust manufacturing to a qualified drug manufacturer. In the entrusted-production case, the holder and the manufacturer must sign both an entrustment agreement and a quality agreement, and they must perform the obligations specified in those agreements (National Medical Products Administration, 2019). Subsequent manufacturing supervision rules sharpen the operational meaning of this arrangement by requiring quality-assurance assessment, risk-management assessment, agreements, and supervision of performance (National Medical Products Administration, 2022b, 2023). In economic terms, the delegated-production route is not converted into a casual outsourcing margin. It is converted into a standardized, legally supported, and supervised route.

A.2 Holder responsibility after production is separated

For the economics of the paper, the important institutional point is that MAH separates production performance from ultimate holder responsibility. It does not transfer the core accountability for the drug away from the holder. Once a drug is authorized, the holder remains the focal entity for the drug across the life cycle.

At a minimum, four dimensions remain holder-side responsibilities:

1. **Lifecycle responsibility.** The holder remains the legally central entity from research and registration through production, distribution, post-marketing study, and pharmacovigilance.
2. **Quality-system responsibility.** The holder must establish and maintain a quality assurance system rather than rely passively on the contractor's internal systems.
3. **Oversight responsibility.** The holder must audit and supervise the quality-management systems and performance of entrusted manufacturers and other entrusted parties.
4. **Safety and release responsibility.** The holder remains the entity around which the law and supervisory rules organize quality, safety, and effectiveness responsibilities, even when physical production is performed elsewhere.

This institutional structure explains why the entrusted route is economically subtle. MAH did not create a zero-cost outsourcing technology. A research-originating holder cannot simply hand off production and walk away. In the model, this residual holder-side burden is captured by μ_i^E. MAH can lower the policy and contractual friction τ_i^E of using the retained external route, but it does not eliminate monitoring, quality-control, technical-transfer, or coordination burdens.

A.3 What the entrusted-route friction is not

The institutional record also clarifies what τ^E and μ^E are not.

Not a demand shock. The legal core of MAH concerns who may hold authorization, who may manufacture, how entrusted production is governed, and who bears quality responsibility. These are organizational and regulatory-allocation changes. They do not by themselves change reimbursement rules, final demand conditions, or therapeutic demand for a drug. Demand-side changes may coexist in the data through other policies, but they are not the distinctive institutional content of MAH.

This does not mean that market demand is absent from the model. Demand-side conditions enter through the single reduced-form market-return shifter R_t in equation (9). The key boundary is that R_t is a background payoff shifter and calibration control, while MAH operates through commercialization-route feasibility and the relative attractiveness of the retained external route.

Not a scientific-productivity shock. Nothing in the MAH architecture changes the scientific technology by which an idea is discovered. The reform may raise expected payoff to research because more ideas become commercializable or because a route becomes more feasible, but that is an expected-return channel, not a direct productivity shift in the scientific production function.

Not a generic fixed-cost reduction for all firms. MAH does not simply relax every firm's operating burden. The post-law architecture may impose more explicit documentation, auditing, and quality-system obligations on the holder. What changes is that these obligations are attached to a legally recognized and standardized external route, so that the MAH-reducible component τ_i^E can fall relative to the pre-reform environment. The residual component μ_i^E remains in the entrusted-route payoff. In other words, MAH can preserve or sharpen holder-side responsibility while still lowering the policy and contractual cost of external implementation by making the arrangement legally supported, contractually structured, and regulatorily intelligible.

This is why the model separates p_m, τ_i^E, and μ_i^E. The direct service price p_m is distinct from the MAH-reducible policy friction τ_i^E, and both are distinct from residual holder-side monitoring and quality-control burden μ_i^E. The comparative statics in the model should therefore be read as statements about the private option value of a legally recognized entrusted-production route, not as statements about demand, science, or a generic deregulation shock.

B Optional Extension: Contract-Manufacturing Service Market

The baseline model takes the contract-manufacturing service price p_m as given. A natural extension endogenizes p_m through a qualified contract-manufacturing service market.

The demand for entrusted manufacturing services is

$$D_m(p_m; M) = \int \lambda_i^*(M, p_m, \tau^E, \mu^E) P_i(r = E \mid M, p_m, \tau^E, \mu^E) \mathrm{d}H_i, \tag{89}$$

where $\lambda_i^*(M, p_m, \tau^E, \mu^E)$ is the model-implied expected number of successful original-drug opportunities and $P_i(r = E \mid M, p_m, \tau^E, \mu^E)$ is the probability that the entrusted route is chosen.

The supply of qualified contract-manufacturing services is

$$S_m(p_m) = \int s_j(p_m; z_j^C) \mathrm{d}H_C(j), \tag{90}$$

where z_j^C indexes producer-side manufacturing capability.

The service-market-clearing condition is

$$D_m(p_m; M) = S_m(p_m). \tag{91}$$

This extension is not part of the baseline model. It is useful for studying how MAH-induced route demand may affect the contract-manufacturing price p_m. A full general-equilibrium model would additionally require product-market clearing, R&D input-market clearing, free entry, exit, and an invariant firm distribution.

C Overview of Calibration Strategy

This appendix documents the mechanism-calibration blueprint behind Section 8. The objective is not to turn the theory section into a full dynamic structural estimation exercise, nor to claim a complete causal empirical design. The objective is narrower: to use a small set of empirical moments that are close to the mechanism to discipline a small set of composite model objects.

The calibration targets three objects:

(i) the effective entrusted-production wedge $(\tau^E + \mu^E)/\sigma_r$ relative to the route-choice scale;

(ii) the composite scale that converts R&D intensity into original-drug opportunities;

(iii) the distribution of research-oriented entry costs, kept as a future or sensitivity module when the current data do not contain a firm-entry panel.

Demand-side controls discipline the reduced-form market-return background R_t. They are not a fourth MAH mechanism and should not be interpreted as evidence that MAH increases demand.

Therefore, Section 8 should be read as a **mechanism calibration blueprint**, not as full structural identification. It asks whether observed route-use patterns and realized original-drug outcomes are consistent with the mechanism that MAH expanded the feasible commercialization route set and lowered the MAH-reducible policy and contractual friction of the external commercialization route. It does not claim to separately identify legal compliance costs, search costs, residual monitoring burdens, quality-management costs, contract-manufacturing prices, latent idea arrivals, or firm-specific project success probabilities.

D Definition of Moments

A moment is a statistic that can be computed both in the data and in the model. For example, in the data one may compute

$$s_{post}^{E,data} = \frac{\#\{\text{post-reform original-drug records with different holder and producer}\}}{\#\{\text{comparable post-reform original-drug records}\}}.$$

Given parameters, the model can generate the corresponding object:

$$s_{post}^{E,model}(\theta).$$

The calibration logic is to choose parameters θ such that

$$m^{model}(\theta) \approx m^{data}.$$

The moment m can be a route-use share, an original-drug count, an entrant count, an original-drug share, or a holder–producer split. A moment is not a new structural variable. It is the empirical bridge between the model and the data.

E Calibration Blocks

The three calibration blocks in Section 8 are summarized in Table 3.

Table 3: Three Calibration Blocks

Calibration block	Data moment	Model moment	Disciplined object
Route-use data	Post-MAH holder–producer split share; exposure-gradient in split shares; pseudo-count sensitivity only	$\log \frac{s^E_{\ell,post}}{1-s^E_{\ell,post}} = \eta_0 + \eta_1 Exposure_\ell + u_\ell$	Effective entrusted-route wedge or exposure-gradient in that wedge
Original-drug counts	Baseline original-drug approvals, realized products, or applications	$\bar{\lambda}_0^O = \mathbb{E}\left[\frac{\beta a^2}{\kappa}\Gamma^0(a, k, S, \tau^T, \mu^E)\right]$	Composite innovation-arrival scale $\beta a^2/\kappa$
Entrant counts	Research-oriented entrant counts or proxies	$N_E(M) = \int F_e(A)dG_N$	Entry-cost distribution F_e

These blocks correspond to the same mechanism chain:

$$\tau^E \downarrow \Rightarrow \Gamma \uparrow \Rightarrow x^* \uparrow \Rightarrow \lambda^* \uparrow \Rightarrow \Lambda^{agg} \uparrow \Rightarrow N_E \uparrow .$$

Here τ^E is the MAH-reducible policy and contractual friction of the external commercialization route, μ^E is the residual holder-side monitoring and quality-control burden, Γ is the expected commercialization value of a successful original-drug opportunity, x^* is optimal original-drug R&D intensity, λ^* is the arrival rate of original-drug opportunities, Λ^{agg} is aggregate expected original-drug opportunities, and N_E is research-oriented entry.

F Route-Use Moment and Entrusted-Production Wedge

F.1 The Data Moment

The central data object for route use is the holder–producer split. For product record p, define

$$Split_p = \mathbb{I}\{Holder_p \neq Producer_p\}.$$

Restricting attention to original-drug records, define

$$s_t^{E,data} = \frac{\sum_{p\in t} Split_p O_p}{\sum_{p\in t} O_p},$$

where $O_p = 1$ if product p is classified as an original drug or satisfies the paper's baseline original-drug proxy. This share is not the policy treatment. It is an observed outcome that proxies the use of an external commercialization route while retaining authorization.

F.2 Why Route Share Maps to the Entrusted-Route Wedge

After the reform, when the entrusted route is feasible, a firm compares commercialization routes after a successful original-drug opportunity is realized. To isolate the logic, consider only internal production and entrusted production among retained products:

$$G_t^I = R_t - C^I(k) + v,$$

$$G_t^E = R_t - p_{m,t} - \tau_t^E - \mu^E + v.$$

Subtracting internal production from entrusted production gives

$$G_t^E - G_t^I = C^I(k) - p_{m,t} - \tau_t^E - \mu^E.$$

Three points are important. First, R_t cancels because both routes commercialize the drug. Second, v cancels because both routes allow the innovator to retain the original-drug asset. Third, the remaining difference is internal production cost, direct entrusted-manufacturing service price, MAH-reducible route friction, and residual holder-side burden. Thus, a lower τ_t^E makes entrusted production more attractive relative to internal production, holding $p_{m,t}$ and μ^E fixed.

For calibration only, suppose this deterministic route choice is smoothed by a binary logit:

$$s_t^E = \frac{\exp(G_t^E/\sigma_r)}{\exp(G_t^E/\sigma_r) + \exp(G_t^I/\sigma_r)}.$$

Then the odds ratio is

$$\frac{s_t^E}{1 - s_t^E} = \exp\left(\frac{G_t^E - G_t^I}{\sigma_r}\right),$$

and taking logs yields

$$\log\left(\frac{s_t^E}{1 - s_t^E}\right) = \frac{G_t^E - G_t^I}{\sigma_r} = \frac{C^I(k) - p_{m,t} - \tau_t^E - \mu^E}{\sigma_r}.$$

This equation does not say that the route-use share equals τ^E or μ^E. It says that the route-use share, through a retained-product route-choice equation, disciplines the relative attractiveness of the entrusted route. Without additional data on internal production costs, direct contract-manufacturing costs, quality agreements, audit costs, contract terms, and firm-level regulatory execution, the calibration recovers only a composite effective wedge relative to the route-choice scale.

F.3 When a Pre–Post Log-Odds Moment Is Valid

A pre–post log-odds moment is valid only if the pre-reform data contain a legally and economically comparable entrusted-retention route. Under that condition, and if $C^I(k)$, p_m, and μ^E are stable across the comparison or handled through comparable samples and controls, one can write

$$\frac{\Delta\tau}{\sigma_r} = \log\left(\frac{s_{post}^E}{1 - s_{post}^E}\right) - \log\left(\frac{s_{pre}^E}{1 - s_{pre}^E}\right),$$

where

$$\Delta\tau = \tau_{pre}^E - \tau_{post}^E > 0.$$

This is not the baseline interpretation when the MAH-style entrusted route was institutionally unavailable before the reform.

F.4 What If Pre-Reform Route-Use Data Are Unavailable?

This is the central practical issue. If the MAH-style entrusted route was institutionally unavailable before the reform, then there is no comparable pre-reform entrusted-route share. Writing

$$s_{pre}^E = 0$$

as though it were an observed route-use rate is misleading, because

$$\log\left(\frac{s_{pre}^E}{1 - s_{pre}^E}\right) = \log(0)$$

is undefined. The formula cannot be used mechanically. There are three disciplined responses.

Response 1: Treat the pre-reform regime as a route-set-expansion boundary case. Before MAH,

$$\mathcal{R}(0) = \{I, T, A\}.$$

After MAH,

$$\mathcal{R}(1) = \{I, E, T, A\}.$$

Thus there is no comparable pre-reform entrusted-route share. The baseline interpretation should emphasize that MAH first expands the feasible route set and then, conditional on route feasibility, lowers the MAH-reducible policy and contractual friction of the route.

Response 2: Use post-MAH exposure or implementation-intensity variation. If the route is unavailable before MAH, the preferred empirical calibration margin is post-MAH variation across regions, dosage forms, therapeutic classes, or firm types. For example, regions or dosage-form segments with greater pre-reform qualified manufacturing capacity may experience a larger effective decline in external commercialization friction. A transparent reduced-form calibration moment is

$$\log\left(\frac{s^E_{\ell,post}}{1 - s^E_{\ell,post}}\right) = \eta_0 + \eta_1 Exposure_\ell + u_\ell,$$

where ℓ indexes an exposure cell, such as a region, dosage-form segment, firm type, or other implementation-intensity cell. This disciplines relative differences in the effective entrusted-route wedge without imposing a nonexistent pre-reform route-use rate.

Response 3: Use a small pseudo-count only as a sensitivity device. For example, use

$$\tilde{s}^E_{pre} = \frac{0.5}{N_{pre} + 1}$$

instead of zero. This makes the log-odds expression computable, but the pseudo-share should not be interpreted as a real pre-reform route-use rate. It should be reported only as a sensitivity check, not as the baseline calibration moment.

F.5 What This Block Can and Cannot Identify

This block can support the following statement: if the holder–producer split among original-drug records rises after MAH, or if higher-exposure segments show higher post-MAH split shares, and this pattern cannot be explained by changes in product composition, therapeutic class, region, or firm capability composition, the pattern is consistent with a lower effective entrusted-route wedge.

It cannot support the following stronger statement: the data separately identify legal compliance costs, search costs, quality-agreement costs, contracting costs, and contract-manufacturing prices. These components enter the route-choice payoff as a composite wedge. Without additional data on those objects, they cannot be separately recovered.

G Original-Drug Counts and R&D Arrival Scale

G.1 The Model Equation

The firm chooses original-drug R&D intensity x_i. The arrival rate of original-drug opportunities is

$$\lambda_i(x_i) = a_i x_i.$$

Given the expected commercialization value Γ_i^M of a successful opportunity, the first-order condition gives

$$x_i^*(M) = \frac{\beta a_i}{\kappa}\Gamma_i^M.$$

Therefore, the optimal arrival rate is

$$\lambda_i^*(M) = a_i x_i^*(M) = \frac{\beta a_i^2}{\kappa}\Gamma_i^M.$$

Taking expectations in the pre-reform or baseline regime gives

$$\bar{\lambda}_0^O = \mathbb{E}\left[\frac{\beta a^2}{\kappa}\Gamma^0(a, k, S, \tau^T, \mu^E)\right].$$

G.2 The Data Moment

The most direct moment is the baseline number of original-drug outcomes:

$$\bar{\lambda}_0^{O,data} = \frac{\#\{\text{pre-reform original-drug records}\}}{\#\{\text{pre-reform years}\}}.$$

At the firm-year level, define

$$Y_{it}^{O,base} = \sum_{p\in(i,t)} O_p^{base}.$$

If the current dataset is an approval-side realized-product file, this object should be interpreted as realized original-drug products, not latent idea arrival. If future data include CDE applications, IND/CTA records, or clinical-trial registrations, those objects can be used to construct moments closer to upstream innovation-opportunity arrival.

G.3 Why This Moment Only Disciplines a Composite Parameter

From

$$\lambda_i^* = \frac{\beta a_i^2}{\kappa}\Gamma_i,$$

original-drug counts depend on three objects:

(i) a_i: research ability;

(ii) κ: the marginal cost scale of R&D effort;

(iii) Γ_i: the commercialization value of a successful opportunity.

If the data only reveal realized original-drug counts, the model cannot separately identify a_i, κ, and Γ_i. High original-drug counts may reflect strong scientific capability, low R&D effort costs, or high commercialization value. Without project-level R&D costs, project success rates, project values, or rich firm-capability data, these objects cannot be separately recovered.

The disciplined approach is to normalize one object. For example,

$$\kappa = 1,$$

or for a benchmark firm,

$$\Gamma^0(a, k, S, \tau^T, \mu^E) = 1.$$

Then

$$\frac{\beta a^2}{\kappa}$$

is interpreted as a composite innovation-arrival scale.

G.4 What If Pre-Reform Original-Drug Counts Are Sparse?

Sparse pre-reform approvals do not invalidate the mechanism. Drug development cycles are long, and final approvals are downstream outcomes. Pre-reform approval counts can therefore be noisy low-frequency moments. Disciplined responses include:

1. using multi-year averages rather than year-by-year moments;
2. using IND/CTA records, clinical-trial registrations, CDE acceptance numbers, or original-drug applications as upstream proxies;
3. separating realized approvals from upstream applications as distinct moments;
4. reporting robustness under alternative original-drug definitions, such as broad new-drug definitions, strict class-I innovative drugs, or inclusion/exclusion of improved new drugs.

The writing boundary is simple: approval-side data calibrate realized original-drug product counts; application-side data are closer to innovation-opportunity arrival.

H Entry Counts and Entry-Cost Distribution

H.1 The Model Equation

A potential entrant enters if post-entry value exceeds entry cost:

$$A(a, k, S, \tau^T, \mu^E, M) \geq f_e.$$

The post-entry value is

$$A(a, k, S, \tau^T, \mu^E, M) = \frac{\beta^2 a^2}{2\kappa(1-\beta)}[\Gamma^M(a, k, S, \tau^T, \mu^E)]^2.$$

If f_e has conditional distribution function $F_e(\cdot \mid a, k, S, \tau^T, \mu^E)$, the entry probability is

$$\Pr(f_e \leq A(a, k, S, \tau^T, \mu^E, M) \mid a, k, S, \tau^T, \mu^E) = F_e(A(a, k, S, \tau^T, \mu^E, M) \mid a, k, S, \tau^T, \mu^E).$$

Aggregating over potential entrants gives

$$N_E(M) = \int F_e(A(a, k, S, \tau^T, \mu^E, M) \mid a, k, S, \tau^T, \mu^E) dG_N(a, k, S, \tau^T, \mu^E).$$

For a fixed potential entrant, the strict entry-response region is

$$\begin{aligned} f_i^e \in [&A(a_i, k_i, S_i, \tau_i^T, \mu_i^E, 0), \\ &A(a_i, k_i, S_i, \tau_i^T, \mu_i^E, 1)). \end{aligned}$$

This threshold statement mirrors the firm-level route-choice result: MAH changes entry only for potential entrants whose entry costs are between the pre-MAH and post-MAH values.

H.2 The Data Moment

The ideal data moment is the number of research-oriented entrants, not the number of all newly registered pharmaceutical firms. Examples include:

- firms that appear for the first time in original-drug applications;
- firms that sponsor clinical trials for the first time;
- firms that first appear as MAH/holder for original-drug records;
- firms without traditional production capacity that enter original-drug R&D or com-

mercialization records.

Define

$$Entry_{it}^{R} = 1$$

if firm i first enters observable original-drug R&D or original-drug realization activity in year t. Then the data moment is

$$N_{E,t}^{data} = \sum_{i} Entry_{it}^{R}.$$

H.3 Why This Step Should Be Sensitivity or a Future Module

A current approval-side realized-product file is typically not a firm-entry panel. Observing that a firm first appears in an approval record is not equivalent to observing true firm entry or research-oriented entry. There are three reasons:

1. the firm may have been founded long before its first approval;
2. the firm may have upstream R&D activity that has not yet reached final approval;
3. name standardization, acquisitions, subsidiaries, and holder-entity changes can affect the first observed year.

Therefore, entrant-count calibration should not be a baseline result in the current data environment. It should be kept as a future data module or as a sensitivity appendix. A serious calibration of F_e requires a merged firm-year entry panel, business registration data, CDE application data, and firm-capability data.

H.4 How to Calibrate F_e

One can assume a simple one-parameter distribution. For example,

$$f_e \sim \text{Exponential}(\mu),$$

so that

$$F_e(A) = 1 - \exp(-A/\mu).$$

The pre-MAH entry rate can be used to match μ, and the post-MAH change in Γ^M can then be used to predict the post-MAH entry response. A lognormal distribution is also possible, but if the data are thin, it is better not to introduce too many shape parameters.

The interpretation should be modest: this step does not estimate each potential entrant's true entry cost. It checks whether the entry margin moves in the direction predicted by a lower external commercialization wedge.

I Handling Missing Pre-Reform Data

Missing or weak pre-reform data are not a minor nuisance. They are central to the calibration design. The empirical architecture should therefore be modular.

I.1 Module A: Current Approval-Side Realized-Product Moments

The most credible baseline moments are:

1. realized original-drug product counts;
2. original-drug share among realized products;
3. holder–producer split;
4. split-based route-use proxy among original-drug records.

These objects can be constructed from the current approval-side matching file, but they are realized-product objects rather than full R&D-process objects.

I.2 Module B: Future Application and Status Panel

Once CDE applications, IND/CTA records, clinical trials, withdrawn projects, pending projects, rejected projects, and approved projects are merged, one can construct:

1. application-to-approval conversion;
2. project realization probabilities;
3. upstream original-drug opportunity arrival;
4. delay and conversion moments.

These moments are closer to λ_i and commercialization success.

I.3 Module C: Future Firm-Capability and Entry Panel

Once business-registration records, production licenses, GMP or dosage-form scope, R&D expenditure, patents, and clinical-trial sponsorship are merged, one can construct:

1. pre-reform research capability;
2. pre-reform manufacturing capability;
3. research-oriented entry;
4. capability-bin-specific route-use and realization moments.

These moments are needed to calibrate the firm-characteristic distribution, F_e, and heterogeneous responses more credibly.

J Recommended Scope for Appendix

The following scope limits should be explicit after Section 8:

1. **What can be claimed:** route-use proxies and realized original-drug outcomes can discipline whether observed patterns are consistent with route-set expansion and a lower composite effective entrusted-production wedge.
2. **What cannot be claimed:** the current data do not separately estimate contract-manufacturing prices, legal compliance costs, firm-specific project success probabilities, latent idea arrivals, or firm-specific true governance costs.
3. **Current baseline:** use approval-side realized-product moments for mechanism calibration; do not treat approval-side data as a full R&D-process panel.
4. **Future upgrade:** once application panels, firm-year entry panels, and capability panels are merged, the exercise can be expanded toward a fuller SMM or indirect-inference design.

K Suggested Text for Main Paper

The following paragraph can be used directly in the appendix or adapted for the main text:

> The calibration strategy in Section 8 should be interpreted as a mechanism-calibration blueprint rather than full structural estimation. The route-use block disciplines the composite effective route wedge relative to the route-choice scale, including the MAH-reducible policy friction and any residual holder-side burden that cannot be separately observed. The baseline original-drug count block disciplines the composite scale that converts R&D effort into realized original-drug outcomes; and the research-oriented entry block, when suitable data become available, disciplines the entry-cost distribution. Because the MAH-style entrusted route may be structurally unavailable before the reform, the pre-reform split share should not be mechanically inserted into a log-odds formula. The baseline interpretation treats the reform as route-set expansion plus conditional friction reduction. If the data permit, post-MAH exposure or implementation-intensity variation can discipline relative differences in the effective route wedge; pseudo-counts for a zero pre-MAH split share should be reported only as sensitivity devices. The current approval-side data can support realized original-drug counts and holder–producer split, but they do not directly identify latent idea arrival, project-level success probabilities, residual holder-side burden, or true research-oriented entry. The calibration exercise is therefore a disciplined mechanism calibration, not full structural identification.

L Appendix Conclusion

The correct interpretation of Section 8 is that data moments discipline composite model objects rather than every structural primitive. Route-use data discipline the effective entrusted-route wedge relative to the route-choice scale only under a retained-product route comparison, a comparable pre-reform route, or post-MAH exposure variation. Original-drug counts discipline the composite innovation-arrival scale $\beta a^2/\kappa$, and research-oriented entry disciplines F_e only when an appropriate firm-entry panel is available. When many pre-reform data objects are unavailable, the baseline must remain modest: the current exercise is approval-side mechanism calibration, while richer entry and state calibration should be reserved for future merged data modules.